MIL-P-22298A(SH)
17 June 1985
SUPERSEDING
MIL-P-22298(SHIPS)
18 January 1960
(See 6.6)

MILITARY SPECIFICATION

PAINT, BLACK, POLYISOBUTYLENE (FORMULA NO. 133)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers black polyisobutylene paint for use on rubber surfaces and surfaces coated with rubber cement. This paint with polyisobutylene antifouling paint, Navy formula no. 134 forms a paint system for underwater rubber surfaces.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-N-95 - Naphtha, Aliphatic.
TT-T-291 - Thinner, Paint, Mineral Spirits, Regular and Odorless.
PPP-P-1892 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.

Beneficial comments (recommendations, additions; deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8010

DISTRIBUTION STATEMENT A Approved for public release; distribution unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:49Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-141 - Paint, Varnish, Laquer, and Related Materials; Methods for Sampling and Testing.
FED-STD-313 - Material Safety Data Sheets Preparation and the Submission of.
FED-STD-595 - Colors.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 93 - Standard Test Methods for Flash Point by Pensky-Martens Closed Tester. (DoD adopted)
D 562 - Standard Test Methods for Consistency of Paints Using the Stormer Viscometer. (DoD adopted)
D 1208 - Standard Test Methods for Common Properties of Certain Pigments. (DoD adopted)
D 1210 - Standard Test Method for Fineness of Dispersion of Pigment-Vehicle Systems. (DoD adopted)
D 1296 - Standard Test Method for Odor of Volatile Solvents and Diluents. (DoD adopted)
D 1475 - Standard Test Method for Density of Paint, Varnish, Lacquer, and Related Products. (DoD adopted)
D 1483 - Standard Test Method for Oil Absorption of Pigments by Gardner-Coleman Method. (DoD adopted)
D 1545 - Standard Test Method for Viscosity of Transparent Liquids by Bubble Time Method. (DoD adopted)
D 1729 - Standard Practice for Visual Evaluation of Color Differences of Opaque Materials. (DoD adopted)
D 2369 - Standard Test Method for Volatile Content of Coatings. (DoD adopted)
D 2698 - Standard Method for Determination of the Pigment Content of Solvent-Type Paints by High-Speed Centrifuging. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rules and Regulations
Rule 102

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 E. Flair Drive, El Monte, CA 91731.)

Source: https://assist.dla.mil -- Downloaded: 20260-19T11:49Z
Check the source to verify that this is the current version before use.

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Formula. The paint shall consist of ingredients conforming to the applicable specifications in the proportions specified in table I. The formula specified is slightly in excess of 100 gallons to allow for normal manufacturing loss. The formula specified in table I is designated as Navy standard formula no. 133. When formula no. 133 is specified, the paint shall conform to this specification.

TABLE I. Formula no. 133.

Ingredients	Pounds per 100 gallons
Carbon black pigment 1/	29.0
Polyisobutylene polymer 2/	34.0
Mineral spirits (type II of TT-T-291)	618.0
Inhibitor (di-tert-butyl-para-cresol) 3/	0.04

1/ The pigment shall be a furnace type carbon black having an oil absorption of not less than 100 determined in accordance with ASTM D 1483 and a fixed carbon content of not less than 98 percent where fixed carbon content is defined as that percent by weight of the original carbon black sample remaining after 7 minutes in a loosely closed platinum crucible at 1750 degrees Fahrenheit (°F).

2/ The polyisobutylene polymer shall be composed of polymerized isobutylene. A 7.5 percent solution of the polymer in mineral spirits conforming to type II of TT-T-291 shall have a viscosity between Z1 and Z4 on the Gardner-Holdt scale when tested in accordance with ASTM D 1545.

3/ CAO-1 of Ashland Chemical Company, Naugard BHT of Uniroyal, DBPC of Koppers Company, or equivalent.

3.2 Manufacture. The component raw materials of the paint shall be mixed and ground as required to produce a product which is uniform, stable, free from grit, suitable for the purpose intended, and in full conformity to the requirements of this specification. The processes used shall not degrade the polyisobutylene (see 6.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:49Z
Check the source to verify that this is the current version before use.

3.3 Volatile portion. The volatile portion of the enamel shall conform to the following requirements by volume (see 4.4):

(a) Aromatic compounds with eight or more carbon atoms except ethylbenzene: 8 percent maximum.
(b) Ethylbenzene toluene and branched keytones: 20 percent maximum.
(c) Compounds with olefinic or cyclo-olefinic type of unsaturation: 6 percent maximum.
(d) Total of (a), (b) and (c): 20 percent maximum.

3.4 Quantitative requirements. The paint shall conform to table II, when tested in accordance with table III.

TABLE II. Quantitative requirements.

Characteristics	Minimum	Maximum
Pigment, percent by weight	3.9	4.4
Volatiles, percent by weight	90.0	91.5
Nonvolatile vehicle, percent by weight	4.8	5.3
Water, percent by weight	---	0.5
Coarse particles and skins as residue retained on standard no. 325 sieve, percent by weight	---	0.2
Consistency, Krebs-Stormer shearing rate 200 r/min:		
Grams	95.0	165.0
Krebs unit (equivalent)	60.0	75.0
Weight per gallon, pounds	6.6	7.0
Dry hard time, minutes	---	90.0
Ash, percent by weight	---	0.2
Fineness of grind	6.0	---
Flash point	27°C (80°F)	---

3.5 Qualitative requirements. The paint shall conform to qualitative requirements as specified in 3.5.1 through 3.5.6.

3.5.1 Odor. The odor shall be characteristic of the volatiles permitted.

3.5.2 Color. The color shall approximately match color no. 37038 of FED-STD-595.

3.5.3 Surface appearance. The paint applied as specified in 4.3.5.2 shall be smooth and uniform in surface appearance.

3.5.4 Adhesion. The paint shall show adhesion to rubber and resistance to stripping when tested as specified in 4.3.5.2.

3.5.5 Elasticity. The paint shall show no change in film characteristics and adhesion when tested as specified in 4.3.5.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:49Z
Check the source to verify that this is the current version before use.

3.5.6 Storage stability. For at least 1 year after date of acceptance, the paint shall readily break up with a paddle to a smooth, uniform consistency; shall not exceed 100 Krebs units in viscosity; shall not exceed 2 hours dry hard time; shall not curdle, gel, nor show any other objectionable properties such as pigment settling.

3.6 Hazardous materials. Lead, chromate, or asbestos shall not be present.

3.7 Toxicity. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting activity.

3.8 Material safety data sheet. The contracting activity shall be provided a material safety data sheet (MSDS) at the time of contract award. The MSDS is DD Form OSHA 20 and is found in FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance. Quality assurance shall be in accordance with method 1031 of FED-STD-141 and as specified in table III. Submission of a certificate of compliance in accordance with method 1031 shall cover complete tests other than the 1-year storage stability test. The storage stability requirement shall apply for 1 year after manufacture regardless of other testing or prior acceptance of the material.

4.2.1 Inspection lot. A lot shall consist of paint of the same type, class, and grade from a single uniform batch or a uniform blend of batches offered for delivery at one time.

4.2.2 Sampling plan. Two representative random samples from each lot shall be tested. The samples shall be packaged in separate containers. The minimum size for each sample shall be 1 quart.

4.3 Test procedures. The paint shall be tested in accordance with table III. Additional chemical and physical tests shall be run, as necessary, to determine that the ingredients and proportions specified in 3.1 have been used. Failure of any sample to pass any test and noncompliance with the requirements of this specification shall be cause for rejection of the lot represented by the sample.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:49Z
Check the source to verify that this is the current version before use.

TABLE III. Test procedures.

Tests	Applicable test method FED-STD-141	Applicable ASTM test method	Test paragraph	Requirement paragraph
Pigment content	4021	D 2698	4.3.1	Table II
Volatiles	----	D 2369	----	Table II
Nonvolatile vehicle content	4053	----	----	Table II
Water	4081	----	----	Table II
Coarse particles	4092	----	----	Table II
Consistency	----	D 562	----	Table II
Weight per gallon	----	D 1475	----	Table II
Drying time	4061	----	4.3.2	Table II
Ash	----	D 1208	----	Table II
Fineness of grind	----	D 1210	----	Table II
Flash point	----	D 93	----	Table II
Odor	----	D 1296	----	3.5.1
Color	----	D 1729	4.3.3	3.5.2
Surface appearance	----	----	4.3.4	3.5.3
Adhesion	----	----	4.3.5.2	3.5.4
Elasticity	----	----	4.3.5.3	3.5.5
Stability in full container	3011	----	4.3.6	3.5.6

4.3.1 Pigment content. The pigment content of the paint shall be determined in accordance with method 4021 of FED-STD-141 or ASTM D 2698 using toluene as the extraction liquid. If the extraction cannot be accomplished by centrifuging, the fixed carbon content of the paint shall be determined by the following method: Weigh a 2-gram sample accurately into a platinum crucible. Dry the paint in a 220°F oven for 2 hours and then volatilize the polymer in a 500°F oven for 1 hour. Place a loose cover over the crucible and heat the crucible for 7 minutes in a 1750°F oven. Cool carefully in desiccator and weigh accurately. Calculate the percentage of the residue left after heating at 1750°F as pigment content.

4.3.2 Drying time. Drying time shall be determined in accordance with method 4061 of FED-STD-141, except the wet film thickness of paint drawn down shall be 3 mils.

4.3.3 Color. The color of the dry paint film prepared in accordance with 4.3.2 shall be compared with color 37038 of FED-STD-595 in accordance with ASTM D 1729.

4.3.4 Surface appearance. The surface of the dry paint film prepared in accordance with 4.3.5.2 shall be examined for compliance with 3.5.3.

4.3.5 Adhesion and elasticity. Adhesion and elasticity shall be determined as specified in 4.3.5.1 through 4.3.5.3 and with temperature and humidity in accordance with FED-STD-141.

4.3.5.1 Composition of rubber on which the coating is applied shall be as specified in table IV.

Source: https://assist.dla.mil -- Downloaded: 202610-19T11:49Z
Check the source to verify that this is the current version before use.

TABLE IV. Rubber composition.

Materials	Parts by weight
Smoked sheet	100.0
Sulphur	3.0
Zinc oxide	5.0
Captax 1/ or equivalent (2-mercaptobenzothiazole)	0.5
Stearic acid	0.5
Heliozone 2/ or equivalent (blend of petroleum waxes)	1.0
Neozone 1/ or equivalent (phenyl naphthylamine)	1.0
P33 black 2/ or equivalent	0.5

1/ R.T. Vanderbilt Co., 230 Park Avenue, New York, NY 10017.
2/ E.I. duPont de Nemours and Co., Wilmington, DE 19898.

The rubber shall be vulcanized in a 0.075 by 6 by 6 inch mold using a minimum pressure of 500 pounds per square inch (lb/in^2) at a temperature of 290°F for 15 minutes. (Sample of rubber may be obtained from David Taylor Naval Ship Research and Development Center, Annapolis, MD 21402.)

4.3.5.2 Adhesion. One surface of the test rubber shall be cleaned by wiping firmly and briskly with a small cotton wad moistened with aliphatic naphtha conforming to TT-N-95. The cotton wad shall have a dry weight of approximately 0.5 gram. To minimize swelling of the rubber, the excess naphtha shall be pressed from the wad before wiping and no more than five strokes shall be taken over any single spot on the surface. The rubber surface shall not appear wet during wiping except behind the moving wad during each stroke. After cleaning the surface, 30 minutes shall be allowed for the naphtha to evaporate and the test rubber to resume a flat position. Three coats of the black polyisobutylene paint, 1 or 2 mils wet thickness per coat, shall be applied to the test rubber. Three hours minimum and 18 hours maximum shall be allowed for dry time between coats. Twenty hours after the application of the final coat, the adhesion shall be tested by attempting to strip the paint off the rubber with fingernails. The individual pieces removed shall be less than 1/4 square inch in area.

4.3.5.3 Elasticity. A rubber panel applied with paint in accordance with 4.3.5.2 shall be stretched to an additional 20 percent of its original length. The paint shall adhere and show no signs of cracking or alligatoring when examined under five power magnification.

4.3.6 Storage stability. The Government, at its option, may examine the enamel (stored unopened in its original container under normal warehouse conditions) for compliance with 3.5.6 at any time within 1 year of acceptance. The examination shall be made in accordance with method 3011 of FED-STD-141.

4.4 Certification data/reports. The contractor shall certify that each lot of paint meets specification requirements in accordance with the data ordering document included in the contract (see 6.2.2). In addition to the requirements covered by the data ordering document, the certification data/report shall include the following information:

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:49Z
Check the source to verify that this is the current version before use.

(a) Conformance of each lot of ingredient material to table I, stating test results and source, as applicable (see 4.2).
(b) Certification that volatility requirements are met (see 3.3).
(c) Conformance to specification requirements and test results as specified (see 4.3) for finished paint.

4.5 Toxicity. The contractor shall furnish to the contracting activity the toxicological data and formulations required to evaluate the safety of the material for the proposed use.

4.6 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the applicable test methods specified herein.

4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisition.)

5.1 Packaging, packing and marking. The paint shall be packaged, packed, and marked in accordance with PPP-P-1892. The level of packaging shall be level A, B, or C, and the level of packing shall be level A, B, or C, as specified (see 6.2.1). The paint shall be furnished in 1-gallon cans or 5-gallon pails as specified (see 6.2.1).

5.1.1 Special marking. In addition to other marking required on the containers, there shall be the following statement: "The volatile content of this container is not photochemically reactive as defined by Rule 102 of the South Coast Air Quality Management District" (see 6.5).

6. NOTES

6.1 Intended use. Black polyisobutylene paint is intended for use as a barrier coat between copper-containing polyisobutylene antifouling paint, Navy formula no. 134 in accordance with MIL-P-22299, and rubber or rubber cement surfaces sensitive to copper.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Size of container (see 5.1).
(c) Level of packaging and packing required (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 202640-19T11:49Z
Check the source to verify that this is the currentversion before use.

6.2.1.1 Level B packaging. Level B packaging is intended to provide economical but limited protection, and should be specified only when it is determined the paint will be held in covered storage no more than 1 year from the date of initial packaging.

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 52.227-7031 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraph.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.4	Certification data/report	UDI-A-23264	10.2.1

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any tasks in section 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Paint should be purchased by volume; the unit being a U.S. gallon at 60°F (15.5 degrees Celsius (°C)).

6.4 Recommended manufacturing process. The polyisobutylene polymer (cut into small pieces) and inhibitor should be added to nearly all the paint thinner to form a 7.5 percent solution. (Unlike rubber, polyisobutylene polymer should not be masticated before preparing the solution as this causes degradation with resulting deterioration of mechanical properties. If rolls are used to draw the polymer out in thin sheets before introduction into the paint thinner, mastication should be held to the least possible amount. Solution of the polymer is a slow swelling process relatively independent of the rate of agitation. The minimum agitation which prevents settling and compaction of the polymer is recommended.) The cuprous oxide pigment should be mixed with a portion of the 7.5 percent polymer solution to make a mill paste, and the paste ground on a roller mill. The rosin should be dissolved in the rest of the paint thinner and the toluene. The mill paste, remaining polyisobutylene polymer solution, and rosin solution should be blended to form the finished paint.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:49Z
Check the source to verify that this is the current version before use.

6.5 Volatile content. Although the container marking specifically refers to the South Coast Air Quality Management District, the paints may be used anywhere else a product complying with 3.3 is allowed. This includes other air pollution control districts or similar areas controlling the emission of solvents into the atmosphere. Information regarding Rule 102 may be obtained from the South Coast Air Quality Management District.

6.6 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 8010-N270)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:49Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:49Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-22298A(SH)	Paint, Black, Polyisobutylene (Formula No. 133)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:49Z
Check the source to verify that this is the current version before use.